

KARTHIK M

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PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate interested in software development, machine learning, and intelligent systems. Research interests include offline AI on edge devices and Deep Reinforcement Learning, with experience building software solutions for compliance, placement readiness, and library management.

EDUCATION

Vidyavardhaka College of Engineering
B.E. in Computer Science and Engineering

Mysuru, Karnataka
CGPA: 8.51/10

RESEARCH EXPERIENCE

PRISM – Offline Conversational AI

2026

First Author — Research Paper

ICIDTSM 2026, IIT Madras

- First author of a research paper on fully offline conversational AI for resource-constrained edge devices.
- Developed a hybrid architecture integrating contextual reasoning, layered memory, lightweight semantic retrieval, and TinyLlama-1.1B inference.
- Paper accepted for presentation at ICIDTSM 2026, IIT Madras.

Deep Reinforcement Learning Research

2026

Research Paper — PPO-Controlled NPCs

Deep Reinforcement Learning

- Contributed to evaluating Proximal Policy Optimization (PPO) against a Finite State Machine (FSM) baseline for non-player character (NPC) behavior.
- Worked on reward-function components and comparative evaluation metrics for PPO-controlled NPCs.

PROJECTS

MANAKSETU | *Next.js, React, TypeScript, FastAPI, Python, OCR*

[GitHub](#)

- Built an AI-assisted Legal Metrology compliance inspection platform that extracts packaged-product declarations from labels and validates them against configurable rules.
- Designed an OCR-to-compliance workflow with evidence-backed findings, inspector verification, and PDF report generation.

SmartPRS | *Next.js, TypeScript, Supabase, AI*

[GitHub](#)

- Developed an AI-powered placement-readiness platform for engineering students, covering aptitude, coding, core skills, soft skills, and attendance.
- Designed placement-readiness scoring, personalized AI coaching, company eligibility predictions, and placement analytics for students and faculty.

BookSphere | *Next.js, TypeScript, Supabase, PostgreSQL*

[GitHub](#)

- Developed a full-stack college library management system using Next.js, TypeScript, and Supabase, enabling students to search books and submit borrowing requests.
- Implemented role-based student and admin dashboards with authentication, inventory management, borrow-request approvals, and PostgreSQL Row Level Security (RLS).

LEADERSHIP & ACTIVITIES

Cerenetics

Present

Co-Founder — Software Solutions Startup

[LinkedIn](#)

- Co-founded a software solutions startup focused on building custom software systems and modernizing existing business technology.

Hackathons & Technical Competitions

Participant

- Participated in multiple hackathons, including Hack-A-League 4.0, Hack-Olympic, Parivarthan 2026, and Symbiot.

V-Vision Buildathon & Math Penta Fest

Competition Achievements

- Received Best Performance recognition at the V-Vision Buildathon.
- Secured first place at Math Penta Fest for pitching the best idea in the ideathon.

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript, TypeScript, SQL, HTML, CSS

Web / Backend: React, Next.js, FastAPI, Flask

AI / ML: Scikit-learn, TinyLlama, Ollama, OCR, Deep Reinforcement Learning

Databases: PostgreSQL, SQLite, Supabase

Tools: Git, GitHub, Docker, VS Code, IntelliJ IDEA, Raspberry Pi